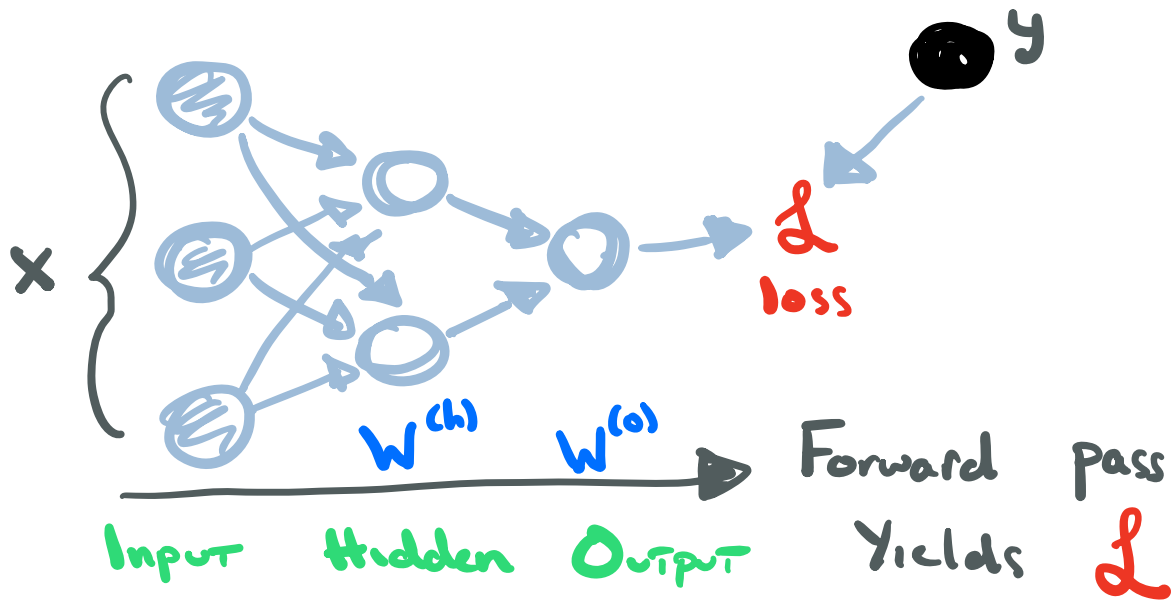


DS 4440 Backpropagation (1)

Last Time Computation graphs comprise layers to form Networks



Today: How to adjust parameters in computation graphs to minimize $\mathcal{L}$.

Backprop for SGD (L1, HW2).

The Recipe

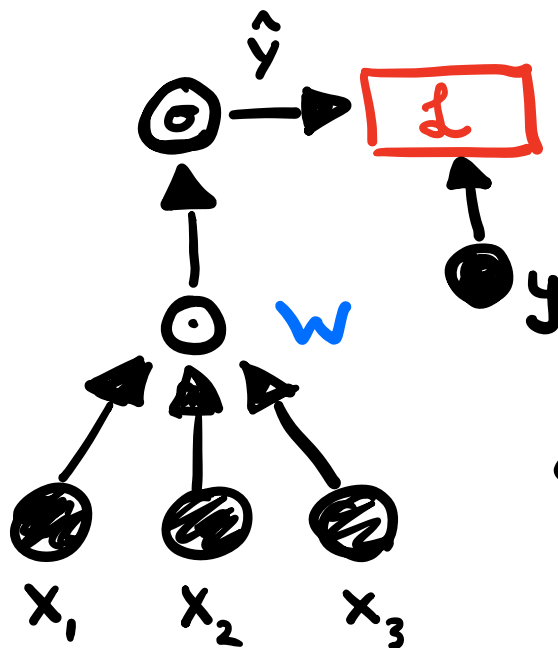
- Build graph (model) w/ parameters W

- Define (differentiable) $\mathcal{L}$
- Use $\nabla_{\mathbf{w}} \mathcal{L}$ to find $\hat{\mathbf{w}}$ via SGD

Problem Finding $\nabla_{\mathbf{w}}$ is painful to do manually for big models.

Solution Use The Computation graph to auto-diff via backprop.

Warm up: logistic regression



$$\hat{y} = \sigma(\underbrace{\mathbf{w} \cdot \mathbf{x}}_{\mathbf{z}})$$

$$\mathcal{L}_{\text{loss}} = -P(y|x, \hat{\mathbf{w}})$$

Start with $\frac{\partial \mathcal{L}}{\partial \hat{y}}$.

$$= -[y \log \hat{y} +$$

$$(1-y) \log (1-\hat{y})]$$

$$\frac{\partial}{\partial \hat{y}} y \log \hat{y} = \frac{y}{\hat{y}}$$

$$u = (1 - \hat{y})$$

$$\begin{aligned} \frac{\partial}{\partial \hat{y}} (1 - y) \log u &= (1 - y) \cdot \frac{1}{u} \frac{\partial u}{\partial \hat{y}} \\ &= - \frac{(1 - y)}{(1 - \hat{y})} \end{aligned}$$

$\underbrace{\frac{\partial u}{\partial \hat{y}}}_{=-1}$

So we have:

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = - \left[\frac{y}{\hat{y}} - \frac{(1 - y)}{(1 - \hat{y})} \right]$$

Two remaining local derivatives:

$$\begin{aligned} \frac{\partial \hat{y}}{\partial z} &= \frac{\partial}{\partial z} \sigma(z) = \sigma(z)(1 - \sigma(z)) \\ &= \hat{y}(1 - \hat{y}) \end{aligned}$$

$$\frac{\partial z}{\partial w} = \frac{\partial}{\partial w} w \cdot x = x$$

Now: ⟨incoming⟩ · ⟨local⟩

$$\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z}$$

$$\begin{aligned}
&= - \left[\frac{y}{\hat{y}} - \frac{(1-y)}{(1-\hat{y})} \right] \hat{y} (1-\hat{y}) \\
&= - [y(1-\hat{y}) - \hat{y}(1-y)] \\
&= - [y - y\hat{y} - \hat{y} + \hat{y}y] \\
&= - [y - \hat{y}] = \hat{y} - y
\end{aligned}$$

Finally:

$$\begin{aligned}
&\frac{\partial \overset{\text{<incoming>}}{\mathcal{L}}}{\partial \overset{\text{<local>}}{w}} = \frac{\partial \overset{\text{<incoming>}}{\mathcal{L}}}{\partial z} \frac{\partial z}{\partial \overset{\text{<local>}}{w}} \\
&= (\hat{y} - y) x
\end{aligned}$$

So

$$\nabla_{\overset{\text{<local>}}{w}} \overset{\text{<incoming>}}{\mathcal{L}} = (\sigma(\overset{\text{<local>}}{w} \cdot x) - y) \cdot x$$

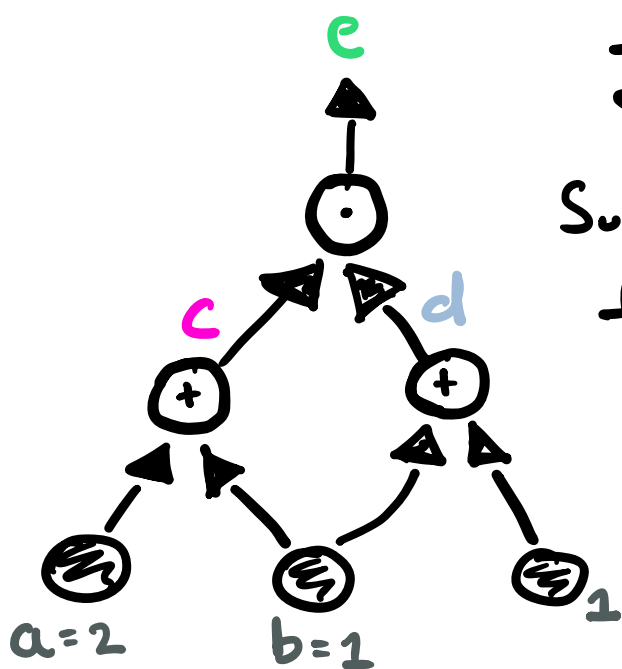
What about Multiple paths?

Consider:

$$\overset{\text{green}}{e} = (a+b)(b+1)$$

c
d

$$\frac{\partial}{\partial b} e = ?$$



Sum over multiple paths
from $b \rightarrow e$

$$\begin{aligned} & \text{incoming} \times \text{local} \\ &= \left(\frac{\partial e}{\partial c} \right) \left(\frac{\partial c}{\partial b} \right) + \left(\frac{\partial e}{\partial d} \right) \left(\frac{\partial d}{\partial b} \right) \\ &= (b+1) + (a+b) = 5 \end{aligned}$$

The Trouble is That Summing
Over all paths doesn't Scale.

Backprop computes one quantity
Per node, which we can re-use.

Let $z^{(l)}$ be pre-activation output
at layer l , so $h^{(l)} = \sigma(z^{(l)})$.

Define **Node Error**

$$\delta^{(l)} = \frac{\partial \mathcal{L}}{\partial z^{(l)}}$$

} How do the
Pre activation
outputs affect
the loss

Then:

incoming
(node) error

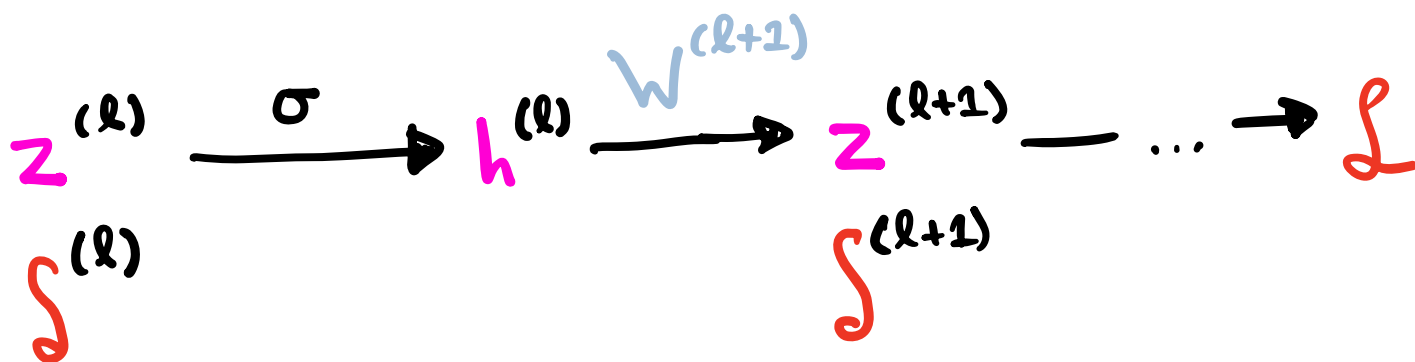
$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}^{(l)}} = \int^{(l)} \cdot (\mathbf{h}^{(l-1)})^T$$

$\langle d_l \times 1 \rangle \langle 1 \times d_{l-1} \rangle$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(l)}} = \int^{(l)}$$

$\langle d_l \times 1 \rangle$

How to compute $\int^{(l)}$?



$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}^{(l)}} = \mathbf{w}^{(l+1)T} \int^{(l+1)}$$

$\langle \text{local} \rangle \cdot \langle \text{incoming} \rangle$

Need to "hop" through σ .

This is element-wise! Consider

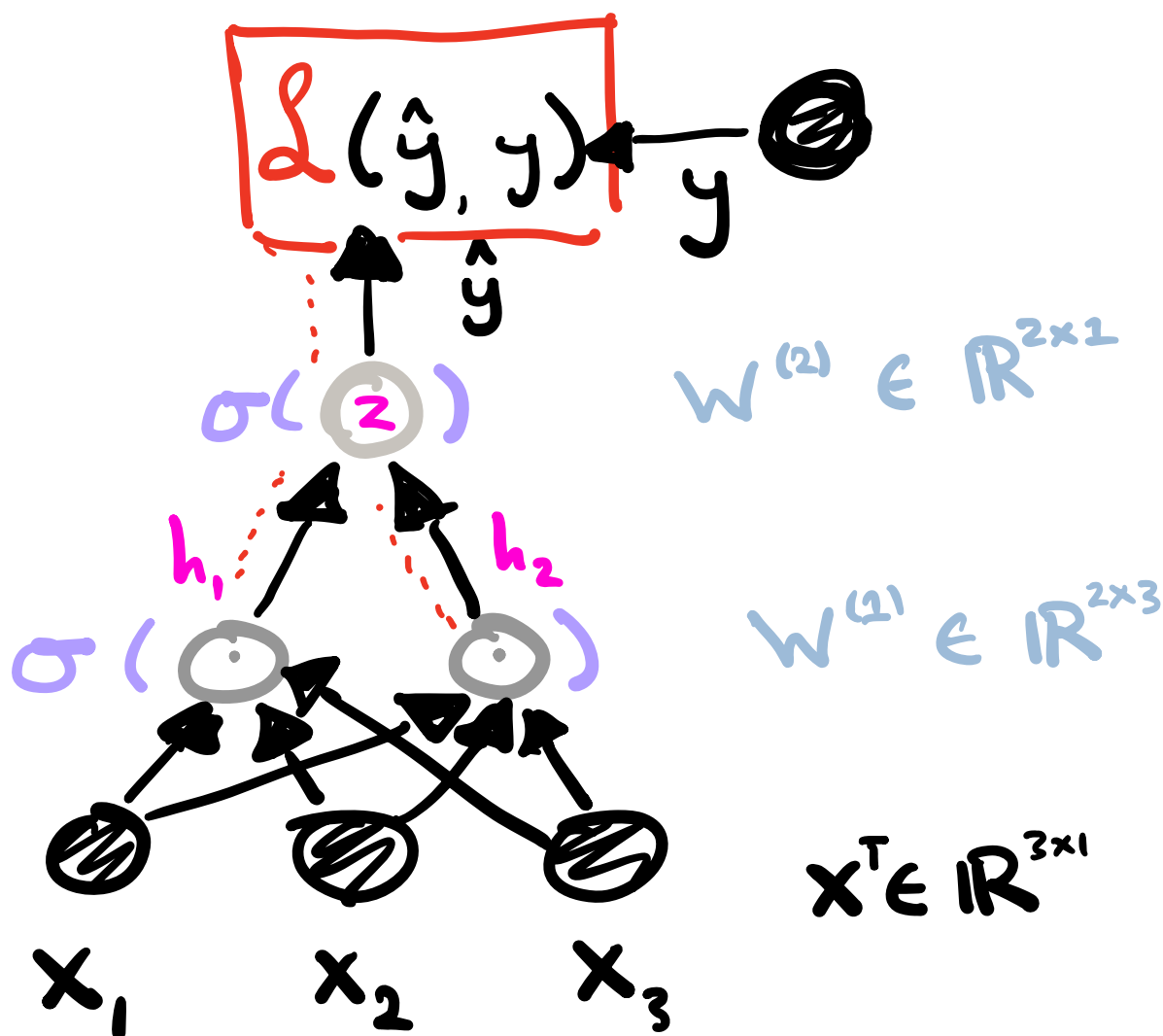
$$\int_i^{(l)} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}_i} = \underbrace{\frac{\partial \mathcal{L}}{\partial \mathbf{h}_i^{(l)}}}_{\text{incoming}} \underbrace{\sigma'(\mathbf{z}_i^{(l)})}_{\text{local}}$$

So in vector form

$$\mathbf{z}^{(l)} = \left(\mathbf{W}^{(l+1)^T} \mathbf{z}^{(l+1)} \right) \odot \sigma'(\mathbf{z}^{(l)})$$

Hadamard Product

Let's work through an example.



Start at the top.

$$\mathbf{z}_{out} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \sigma'(\mathbf{z})$$

logit

Push backwards. We will get

$$\delta_h = \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix} = \begin{bmatrix} \delta_{out} \cdot W_1^{(2)} \sigma'(z_1) \\ \delta_{out} \cdot W_2^{(2)} \sigma'(z_2) \end{bmatrix}$$

Now

$$\Delta_{W^{(1)}} L = \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix} \otimes [x_1 \ x_2 \ x_3]$$

$$= \begin{bmatrix} \delta_1 x_1 & \delta_1 x_2 & \delta_1 x_3 \\ \delta_2 x_1 & \delta_2 x_2 & \delta_2 x_3 \end{bmatrix} = \begin{bmatrix} \delta_1 \vec{x} \\ \delta_2 \vec{x} \end{bmatrix}$$

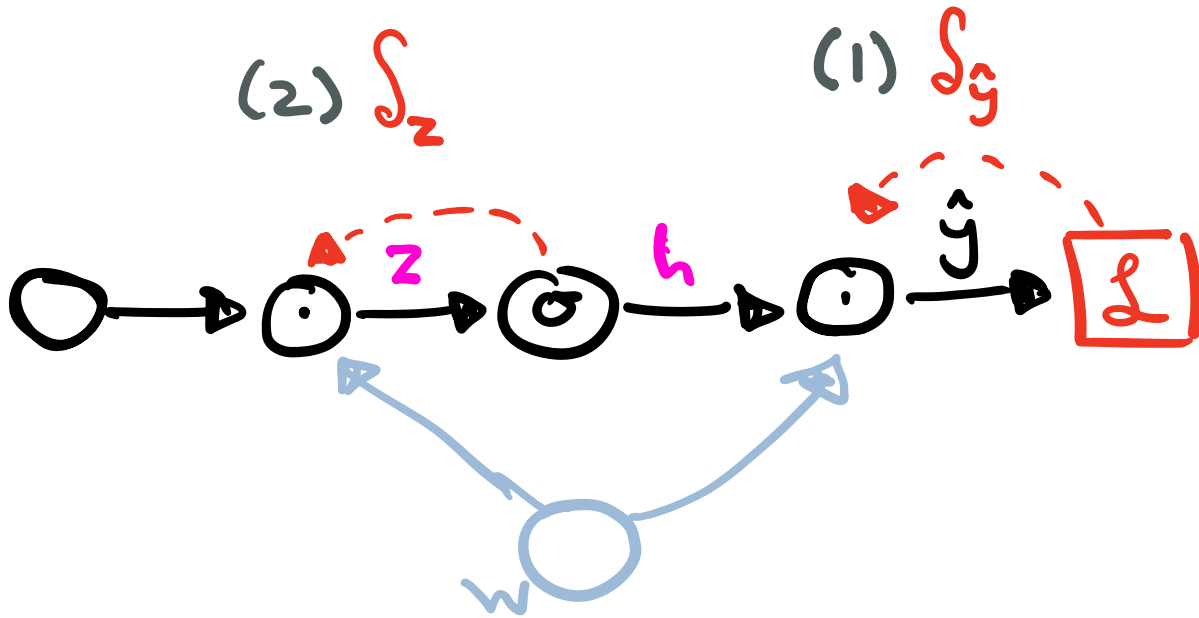
→ Connection from x to h_2
& δ_2 is incoming error There

Some more examples (good quiz prep!)

A Shared Weight

$$\hat{y} = w \sigma(wx) = wh$$

$$h = \sigma(wx) \quad z = wx$$



$$\frac{\partial \mathcal{L}}{\partial w} = \overset{\text{<incoming>}}{\delta_{\hat{y}}} \cdot \overset{\text{<local>}}{h} + \delta_z x$$

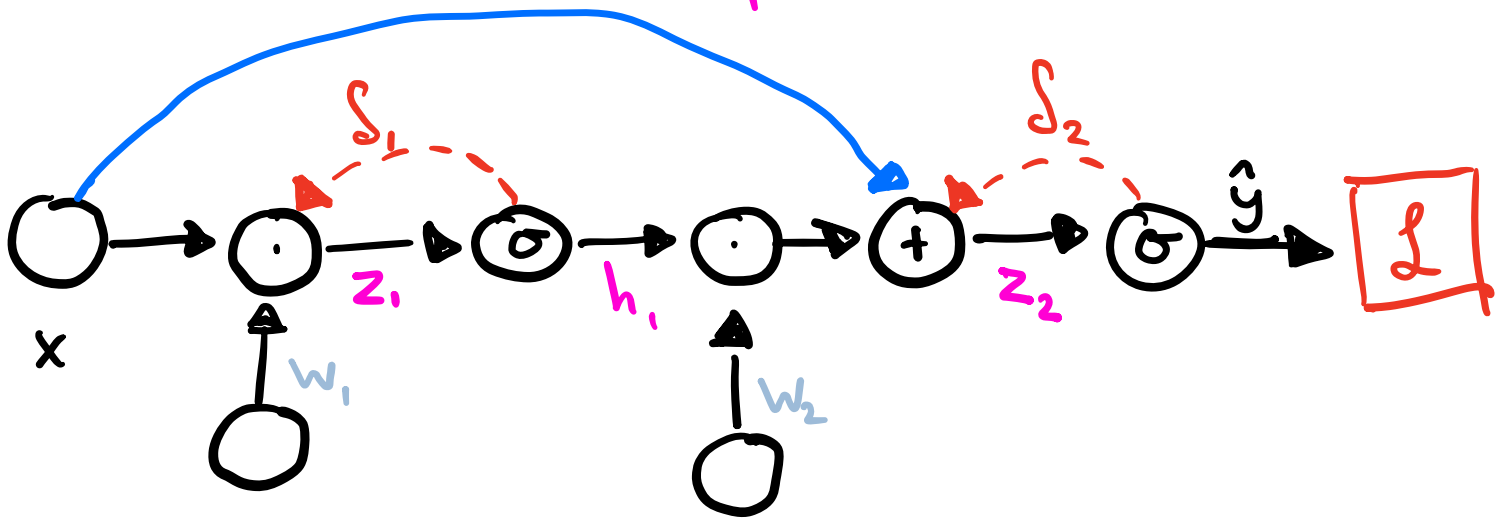
(1) (2)

$$\delta_{\hat{y}} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \quad | \quad \delta_z = \underset{\text{incoming}}{\delta_{\hat{y}}} \cdot \frac{\partial \hat{y}}{\partial h} \cdot \sigma'(z)$$
$$= \delta_{\hat{y}} \cdot w \cdot \sigma'(z)$$

A Skip Connection

$$\hat{y} = \sigma(\underbrace{w_2 h_1 + x}_{z_2})$$

$$h_1 = \sigma(\underbrace{w_1 x}_{z_1})$$



$$\frac{\partial \mathcal{L}}{\partial x} = \delta_1 \frac{\partial z_1}{\partial x} + \delta_2 \frac{\partial z_2}{\partial x}$$

$$= \delta_1 w_1 + \delta_2 (1)$$

$$= \delta_2 \frac{\partial z_2}{\partial h_1} \underbrace{\sigma'(z_1)}_{\text{get to } z_2}$$

$$= \delta_2 w_2 \sigma'(z_1)$$

$$= \delta_2 w_2 \sigma'(z_1) w_1 + \delta_2$$